

Competitor Landscape: Aztec Network (Stage 0)

Stage A – Category Classification

Aztec Network is primarily classified in the following categories (with verification):

- **Privacy-Focused Ethereum Layer-2 (ZK Rollup):** Aztec is a layer-2 rollup on Ethereum that **prioritizes privacy** via zero-knowledge proofs ¹ ². It uses zkSNARKs to enable encrypted smart contracts, making it a “*privacy-focused Ethereum Layer 2 (L2)*” network ².
- **Programmable Privacy / Private Smart Contract Platform:** Unlike simple privacy coins, Aztec offers **fully programmable** private smart contracts (“a privacy-first L2 on Ethereum”) ³. It blends Ethereum scalability with on-chain privacy, essentially a “*private world computer*” for DeFi ⁴.
- **Zero-Knowledge Proof Blockchain Technology:** Aztec leverages advanced **zero-knowledge cryptography** (zkSNARK proofs) at its core ⁵. It is often described as a **ZK-rollup** that integrates privacy, distinguishing it from other ZK-rollups that use validity proofs only for scaling ⁶.

Sources: CoinGecko describes Aztec as a “*privacy-focused Ethereum Layer 2*” ², and CoinDesk notes it as “*a layer-2 rollup focused on privacy,*” emphasizing its zero-knowledge encryption of transactions ¹.

Stage B – Competitor Discovery (Safe)

Below is a table of **validated competitors** to Aztec Network, identified via reputable sources (official docs, aggregators, or media). Each competitor is classified by type, with the reason for inclusion and a verification source. Confidence scores reflect how strongly evidence supports the competition.

Competitor	Type	Reason (≤12 words)	Verification Anchor	Confidence
Aleo	Adjacent	L1 private smart-contract chain (ZK-powered)	“Aleo... aims to make programmable privacy... a standalone Layer 1” ⁷	95
Fhenix	Direct	Ethereum L2 with fully homomorphic encryption privacy	“Fhenix, a forthcoming layer-2... bringing FHE to smart contracts” ⁸	90
Manta Network (Pacific)	Direct	Ethereum L2 for privacy & scalability (ZK + modular)	“Manta Pacific... an EVM L2... combining scalability, privacy” ⁹	85
Polygon Nightfall	Direct	Ethereum L2 rollup for private transactions (EY/ Polygon)	“Aztec... Competitors include Polygon Nightfall...” ¹⁰	80

Competitor	Type	Reason (≤ 12 words)	Verification Anchor	Confidence
Obscuro (TEN)	Direct	Ethereum L2 with TEE-based confidential contracts	"Competitors include... Obscuro, and Intmax. Rollup solutions enable private transactions" ¹¹	80
Intmax	Direct	Ethereum L2 zkRollup with stateless, privacy-centric design	"Competitors include... Obscuro, and Intmax. Rollup solutions enable private transactions" ¹¹	75
StarkNet	Adjacent	ZK-based Ethereum L2 (scaling-focused, no default privacy)	"StarkWare (Starknet)... primarily prioritize scalability (not privacy)" ¹²	70
zkSync Era	Adjacent	ZK-rollup Ethereum L2 (general purpose, no privacy)	"zkSync... primarily prioritize scalability, Aztec focuses on privacy" ¹²	70
Scroll	Adjacent	ZK-EVM Ethereum L2 (focus on EVM compatibility)	"...other L2 solutions such as... Scroll primarily prioritize scalability" ¹²	60
Oasis Network	Adjacent	L1 privacy network with Oasis Privacy Layer for EVM	"Oasis... led the next wave of privacy (confidential smart contracts)" ¹³ ¹⁴	75
Secret Network	Adjacent	L1 for private smart contracts (TEE-based, Cosmos)	"Secret Network... enables privacy-preserving smart contracts using TEEs" ¹⁵	70
Railgun	Adjacent	Ethereum L1 privacy protocol for DeFi (shielded addresses)	"Railgun... integrates privacy into existing DeFi... through shielded addresses" ¹⁶	80
Monero	Narrative	Privacy coin (default anonymous L1, no smart contracts)	"Monero provides a base-layer designed for default privacy" ¹⁶	65
Zcash	Narrative	Privacy coin using zk-SNARKs (payments only)	"privacy-focused blockchains (such as Monero)... limited to simple transfers" ¹⁷	60

- **Aleo** – *Adjacent*: A Layer-1 blockchain for programmable privacy, often positioned alongside Aztec in the "privacy computing" space ⁷. Unlike Aztec's L2 approach, Aleo runs its own chain (Leo language) but shares the goal of fully private dApps ⁷.
- **Fhenix** – *Direct*: An upcoming Ethereum L2 using **Fully Homomorphic Encryption (FHE)** to enable computations on encrypted data ⁸. Targets similar use-cases as Aztec (on-chain confidentiality), but via FHE instead of ZK-SNARKs ¹⁸.
- **Manta Network (Pacific)** – *Direct*: A modular Ethereum L2 that combines ZK proofs with high throughput design ⁹. Aims for scalable private DeFi and interoperates with a Polkadot-based privacy layer (Atlantic) ¹⁹.

- **Polygon Nightfall** – *Direct*: An Ethereum Layer-2 rollup (by Polygon & EY) focused on **private ERC-20 transfers** for enterprises. Uses an Optimistic rollup with zk-proofs for privacy ²⁰ . Mentioned as a peer “privacy rollup” in industry comparisons ¹⁰ .
- **Obscuro (TEN)** – *Direct*: An Ethereum L2 that uses Trusted Execution Environments to keep contract data confidential (a “TEE-based rollup”) ¹⁵ . Listed among Aztec’s direct competitors in privacy rollups ¹¹ . (Obscuro Labs’ product “TEN” raised funding for an encrypted L2 ²¹ .)
- **Intmax** – *Direct*: An Ethereum L2 (stateless zkRollup) focused on privacy and scalability ²² . Often mentioned alongside Aztec as a privacy-enabled rollup solution ¹⁰ . Intmax uses ZK proofs to allow private transfers and claims to require no on-chain data for certain ops (novel design).
- **StarkNet (StarkWare)** – *Adjacent*: A ZK-rollup on Ethereum focusing on **scalability and custom VM (Cairo)**, without default privacy ²³ . Overlaps with Aztec in being an advanced ZK Layer-2, but “is not private...you’d have to build something like Aztec on top of Starknet” to get privacy ²⁴ .
- **zkSync Era** – *Adjacent*: A general-purpose ZK-rollup (EVM-compatible) for scaling Ethereum. It has hinted at future privacy features but currently *uses ZK proofs for scaling, not privacy* (all transactions are public) ²⁵ . Competes as an Ethereum L2 attracting developers, though Aztec’s focus (privacy over throughput) differentiates it ¹² .
- **Scroll** – *Adjacent*: A zkEVM Layer-2 similar to zkSync, aiming for Ethereum equivalence. It prioritizes EVM compatibility and throughput over privacy ¹² . Considered in the same cohort of ZK rollups, but does not provide on-chain confidentiality by default.
- **Oasis Network** – *Adjacent*: A Layer-1 blockchain with a focus on privacy and data tokenization. It offers the **Oasis Privacy Layer** to add confidential execution to other chains ¹⁴ . Oasis is often cited alongside Aztec as part of a “privacy renaissance” in Web3 ¹³ , though Oasis uses trusted hardware (and is not Ethereum-based).
- **Secret Network** – *Adjacent*: A Cosmos-based chain enabling private smart contracts via TEEs. Like Aztec, it supports private logic, but relies on secure enclaves instead of ZKPs ¹⁵ . Secret and Aztec are both examples of extending privacy beyond transactions (e.g. to data and contracts) ²⁶ . Secret isn’t on Ethereum L2, but shares the market of privacy-first platforms.
- **Railgun** – *Adjacent*: An on-chain **privacy protocol for Ethereum** (and other chains) that lets users interact with DeFi apps via shielded addresses ²⁷ . Fulfills similar user needs (private DeFi transactions) by **integrating with existing L1 dApps** ²⁸ . Unlike Aztec’s new rollup chain, Railgun is a smart-contract system (users deposit into a shielded pool and use adaptors for DeFi) ²⁸ . It’s a direct alternative for users seeking Ethereum privacy today.
- **Monero** – *Narrative*: A well-known privacy coin (Layer-1) offering **mandatory private transactions** by design. Not a smart-contract platform, but in the public narrative Monero is a benchmark for financial privacy. Aztec’s approach is often contrasted with Monero/Zcash – Aztec allows DeFi functionality, whereas Monero is limited to payments ¹⁷ .
- **Zcash** – *Narrative*: Another pioneer in blockchain privacy (zk-SNARK-based transactions on an L1 coin). Zcash provides optional shielding for value transfers, but no smart contracts. Aztec aims to achieve what Zcash does for payments *plus* full DeFi programmability ¹⁷ . (Both are mentioned in discussions of privacy tech, though Aztec operates as an Ethereum extension rather than an independent coin.)

Note: We did **not** find credible evidence of some frequently mentioned “competitors” in community chatter. For example, **Optimism/Arbitrum** (optimistic rollups) occasionally come up in comparisons due to being popular Ethereum L2s, but they do *not* offer privacy features – any comparison is about general L2 usage, not privacy, and thus they are not included as true competitors.

Stage C – Uncertain Mentions

These projects have been **floated as possible competitors** in some contexts, but evidence for a direct competition with Aztec is weak or speculative. They are noted here with low confidence:

Project	Why It Is Uncertain	Source of Mention	Confidence
COTI (Layer-2)	Focused on payments & “encrypted computation”; not widely seen as DeFi privacy leader.	COTI News blog listing Aztec as a competitor ²⁹	50
Suterusu (Suter Shield)	Layer-2 privacy protocol (zk-ConSNARK) with low adoption; rarely compared directly.	Slashdot “Aztec alternatives” list (includes Suterusu) ³⁰	30
Polkadot	Web3 interoperability chain; no inherent privacy focus (included in error by some).	Slashdot “Best Aztec Alternatives” misclassified list ³¹	20

- **COTI:** A project pivoting to an Ethereum L2 with **garbled circuits** for privacy. COTI’s own blog acknowledges overlap with Aztec’s goals ²⁹, but its positioning (payments and enterprise use) means it isn’t commonly cited by neutral sources as an Aztec competitor. We list it with caution given the self-published nature of the source.
- **Suterusu:** A protocol that introduced “*Suter Shield*” (a trustless layer-2 privacy solution for transactions) around 2020 ³⁰. It’s conceptually similar (using ZK proofs to hide senders/amounts), but it has minimal market visibility now. It appeared on an algorithmic “alternatives” list, but we found no recent major adoption or direct comparisons involving Aztec.
- **Polkadot:** Included in one automated list of Aztec “alternatives,” likely due to Polkadot’s general prominence ³¹. However, Polkadot is not a privacy-preserving solution *per se*. Any competitive angle is indirect (Polkadot’s ecosystem hosts its own privacy projects like Manta). We consider this an incorrect association rather than a true competitor.

Stage D – Incorrect Comparisons

Finally, we note projects that are **sometimes incorrectly compared** to Aztec Network, along with why those comparisons are misguided:

Project	Why Incorrect (≤12 words)	Example of Misclassification
Tornado Cash	Not a platform, just a mixer (no smart contracts or L2)	<i>FTX flagged Aztec’s zk.money as a “mixer” similar to Tornado</i> ³²
Lightning Network	Bitcoin off-chain scaling, not privacy or Ethereum-based	<i>Slashdot “Aztec Alternatives” inexplicably lists Lightning Network</i> ³³
Arbitrum/Optimism	General L2s (no privacy); scale vs privacy trade-off ignored	<i>No privacy features – comparing only because all are Ethereum L2s (common but incorrect)</i>

- **Tornado Cash:** Aztec Connect’s pool of encrypted funds led some observers (e.g. exchanges) to conflate it with Tornado Cash (an anonymity mixer) ³². This is **incorrect** – Aztec is a full-fledged layer-2 with programmatic privacy, whereas Tornado is a single-purpose mixer for obscuring transfers. The “mixer” label oversimplifies Aztec’s capabilities and was refuted by the team.

- **Lightning Network:** A Bitcoin Layer-2 for fast payments. It was oddly listed as an “Aztec alternative” by one source ³³, which is misleading. Lightning offers *no on-chain privacy* (transactions are just off-chain and eventually public when settled) and is unrelated to Ethereum or smart contracts.
- **Optimism/Arbitrum:** These popular Ethereum L2s often come up in broad L2 comparisons, but any equivalence to Aztec is **misplaced**. They do not provide privacy — in fact, Aztec’s founder noted “*we never need to be as cheap as other L2s*” ³⁴ because Aztec’s value is privacy, not maximal throughput. Thus, treating them as direct competitors ignores Aztec’s unique selling point (privacy). Any competition here is only for general user mindshare on Ethereum scaling, not on privacy features.

Stage E – Pre-Rating Table

Below is a positioning summary for Aztec Network and its key competitors, synthesizing verified insights on each project’s **category**, strengths, weaknesses, current narrative momentum, and risks. (All information is drawn from the above analysis and cited sources.)

Project	Category	Strengths	Weaknesses	Narrative Momentum	Risk Notes	Conf.
Aztec Network	Privacy L2 (zkRollup on Ethereum)	True end-to-end privacy for DeFi ⁶ ; inherits Ethereum security.	Higher tx cost & complexity (privacy proofs limit throughput) ³⁴ .	High – Leading privacy L2, strong recent testnet hype ³⁵ .	Regulatory scrutiny (privacy ≠ illicit use debate) ³⁶ ; heavy proving computation for users ³⁷ .	90
Aleo	Privacy L1 (ZK blockchain)	Full privacy by default; custom VM & language for ZK dApps.	Not Ethereum-compatible (own ecosystem); mainnet delayed ³⁸ .	High – Significant funding & Vitalik-backed interest ³⁹ , but waiting on launch.	Tech complexity for devs (new stack); launch timing uncertainty.	85
Fhenix	Privacy L2 (Ethereum, FHE-based)	Innovative FHE approach (compute on encrypted data) ¹⁸ ; EVM-extension (leverages Ethereum).	Early-stage (testnet pending, unproven at scale) ⁸ ; FHE very computationally heavy.	Moderate – Niche interest from cryptography circles; not widely known yet.	Cutting-edge risk (FHE new in practice); potential performance and UX challenges.	70

Project	Category	Strengths	Weaknesses	Narrative Momentum	Risk Notes	Conf.
Manta Network	Privacy L2 & L1 (Hybrid)	Dual-chain design (Ethereum L2 + Polkadot L1) ⁴⁰ ; ZK-based confidential transactions and identity.	Fragmented ecosystem (split between Polkadot and Ethereum); OP-stack L2 (inherits optimism trade-offs).	Moderate – Gained attention with pivot to Ethereum L2 ⁹ ; not yet a household name.	Still experimental; needs to prove L2 security and privacy efficacy in production.	75
Polygon Nightfall	Privacy L2 (Ethereum, Optimistic)	Backed by EY & Polygon (credibility); aimed at enterprise use (compliance focus).	Optimistic rollout = withdrawal delays; limited to transfers (not general dApps).	Low – Very little recent buzz in retail circles; used mainly in enterprise demos.	Adoption risk (narrow use-case); might be sidelined by more flexible solutions.	60
Obscuro (TEN)	Privacy L2 (Ethereum, TEE-based)	Complete data privacy via TEEs; no changes needed for Solidity devs.	Relies on trusted hardware (Intel SGX); trust and centralization concerns.	Low – Largely under the radar; known in enterprise/blockchain consortia.	Trust assumptions (TEE could be compromised); needs network effect.	60
StarkNet	ZK Rollup L2 (Ethereum, general)	High scalability and custom VM (Cairo) with growing dev community.	No built-in privacy (all transactions public) ²⁴ ; still alpha (limited decentralization).	High – Strong mindshare as a top ZK-rollup (no. of projects deploying).	Execution risk (new language adoption); competition with other L2s in performance.	80
zkSync Era	ZK Rollup L2 (Ethereum, EVM)	EVM-compatible and live today; fast user growth in 2023.	No default privacy (uses ZK only for scaling); had centralization/upgrade keys initially.	High – Significant adoption (DeFi, payments on L2) and ecosystem funding.	Security maturity (new tech bugs – e.g. early issues patched); adding privacy later remains uncertain.	80

Project	Category	Strengths	Weaknesses	Narrative Momentum	Risk Notes	Conf.
Secret Network	Privacy L1 (Cosmos, TEE-based)	Proven mainnet with private smart contracts live; Cosmos interoperability.	Uses TEEs (requires trust in enclaves) ¹⁵ ; not Ethereum-compatible by default.	Moderate – Active in Cosmos niche; less discussed on Ethereum forums.	Regulatory: US-centric cloud hardware dependency; smaller DeFi ecosystem vs Ethereum.	70
Railgun	Privacy protocol (Layer-1 Ethereum)	Deployed on Ethereum & others now; enables DeFi privacy without new chain ²⁸ .	Requires relayers and custom wallet; anonymity set varies by usage (privacy $\approx$ user pool size) ⁴¹ .	Moderate – Steady usage by privacy-seekers; not much mainstream hype.	If usage drops, privacy weakens (needs large user base) ⁴¹ ; subject to similar regulatory risks as mixers.	75
Monero	Privacy Coin (L1, UTXO-based)	Battle-tested privacy (years of obfuscation tech in use); no opt-out of privacy.	No smart contracts or DeFi; siloed ecosystem (only XMR transfers).	Moderate – Continues to be championed for financial privacy, but outside of DeFi boom.	Regulatory delisting risk (seen with exchanges); scalability limits (on-chain only).	65

(All entries derived from earlier tables and sources: e.g., Aztec's strengths/weaknesses from Nansen/Coindesk ⁶ ³⁴ , Aleo's delays ³⁸ , Railgun's anonymity set note ⁴¹ , etc.)

Stage F – Final Question

Would you like to export selected competitors into your Stage-1 Evidence Log prompts?

¹ ³ ⁶ ²⁵ ³⁴ ³⁵ 'Everything Is Encrypted': Aztec's Privacy Rollup Hits Testnet Amid Growing Demand

<https://www.coindesk.com/tech/2025/05/01/everything-is-encrypted-aztec-s-privacy-rollup-hits-testnet-amid-growing-demand>

² ⁵ ¹⁷ ³⁹ What Is Aztec Network (AZTEC)? A Vitalik-backed Privacy ETH L2 | CoinGecko

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⁴ The Privacy-First L2 on Ethereum - Aztec

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